

Topic 1: Introduction

[Summary of J. H. McClellan, R. W. Schafer, and M. A. Yoder, *Signal Processing First*. Prentice Hall, 2003.]

Topics: Signals, systems, continuous-time signals, discrete-time signals, sampling period, mathematical representation, block diagrams, and audio CD recording/reproduction.

Key idea. *Signals carry information through physical pattern variations across time or space, and systems operate on signals to transform them; mathematics provides a precise language to represent, analyze, and design both signals and systems using real-world applications like audio processing.*

1. Purpose and roadmap

Courses in digital signal processing (DSP) generally place considerable emphasis on mathematical techniques and algorithms, and less emphasis on physical signals. This course introduces signal processing through real signals, specifically audio signals. Working with real signals provides stronger motivation for learning DSP theory and practical applications.

The primary objective is to develop a methodology that allows precise description and modeling of both signals and systems. Representing them with mathematical equations clarifies how signals interact with systems and how systems can be designed and implemented to meet given specifications.

Stage	Operation	Main consequence
Signal Generation	<i>Physical variation (e.g., acoustic speech)</i>	Produces a continuous physical signal $s(t)$ carrying information.
Sampling / Conversion	$s[n] = s(nTs)$	Converts continuous-time signals into discrete numerical sequences.
System Processing	$y(t) = T\{x(t)\}$ or $y[n] = T\{x[n]\}$	Manipulates, transforms, or records input signals into new output signals.
Reconstruction	<i>Digital-to-Analog conversion</i>	Reconstructs a continuous-time signal $\hat{x}(t)$ from discrete samples.

2. Signals and their physical representations

Intuitively, a signal can be described as something that carries information. This “something” is usually a pattern of variations in a physical quantity that can be manipulated, stored, and transmitted by a physical process.

An important point is that signals can take many equivalent forms and representations:

- **Acoustic signal:** Speech sound waves propagating through a medium such as air.
- **Electrical signal:** Continuous voltage variations generated by a microphone.
- **Magnetization pattern:** Physical magnetic domain alignment stored on magnetic tape.
- **Numerical sequence:** Discrete binary numbers stored in modern digital audio recordings.

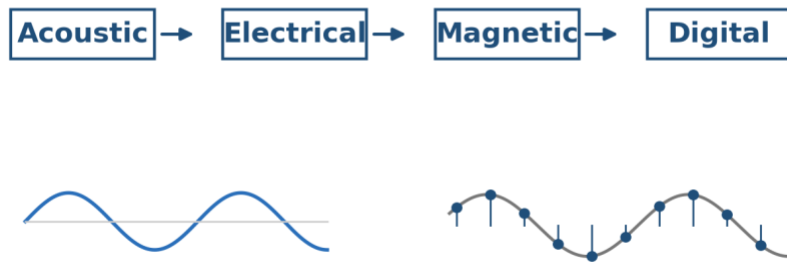


Figure 1.1. Equivalent physical and digital representations of the same signal.

3. Mathematical representation of signals

Signals are patterns of variation that represent or encode information. Mathematics provides a formal framework to represent these variations as functions of independent variables.

3.1 Continuous-time signals

A continuous-time signal is defined for all values of a continuous independent variable, usually time t . It is denoted mathematically as:

$$s(t)$$

Speech voice is a classic example of a one-dimensional (1D) continuous-time signal. When exact algebraic formulas are unavailable, the signal graph itself serves as the definition of the function that assigns a numerical value $s(t)$ to every instant of time t .

Signals can also depend on spatial variables rather than time. For example, a two-dimensional continuous image is represented mathematically as a function of two spatial variables, $s(x, y)$.

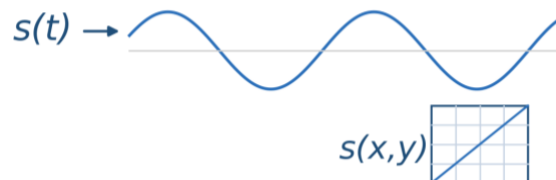


Figure 1.2. Continuous-time signal examples and notation.

3.2 Discrete-time signals

While many physical signals originate as continuous-time processes, digital processing requires discrete representations. This is accomplished by sampling a continuous-time signal at isolated, equally spaced instants.

Mathematically, the sampled sequence is defined as:

$$s[n] = s(n T_s)$$

where n is an integer sample index, $n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$, and T_s is the sampling period (the time interval between consecutive samples).

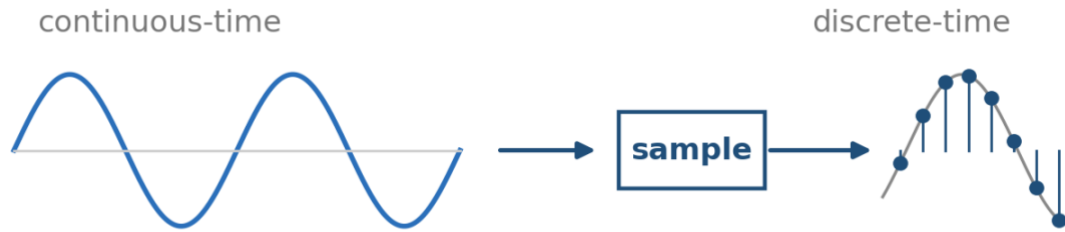


Figure 1.3. Sampling a continuous-time signal to obtain a discrete-time sequence.

4. Mathematical representation of systems

A system is an operator or physical process that transforms an input signal into a new output signal or a different signal representation.

4.1 Continuous and discrete system operators

For a continuous-time system with input signal $x(t)$ and output signal $y(t)$, the transformation is expressed as:

$$y(t) = T \{ x(t) \}$$

A specific example is a magnitude-squaring continuous system:

$$y(t) = x^2(t)$$

If both input and output are discrete-time sequences, the system transformation is written as:

$$y[n] = T \{ x[n] \} \Rightarrow y[n] = x^2[n]$$

Implementing such a discrete-time squarer system in software is simple: each numerical input sample $x[n]$ is multiplied by itself to generate $y[n]$.



Figure 1.4. A system transforms an input signal into an output signal.

4.2 The sampler system

A continuous-to-discrete converter (sampler) is a system whose input is a continuous-time signal $x(t)$ and output is a discrete sequence $x[n]$:

$$x[n] = x(n T_s)$$

The sampler takes instantaneous snapshots of the continuous signal once every T_s seconds.

5. System representation via block diagrams

Engineers use block diagrams as visual representations to show system operations, signal paths, and processing sequences. A block diagram illustrates how signals flow through sub-systems and functional blocks.

5.1 Basic system block diagrams

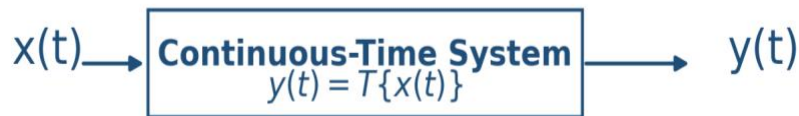


Figure 1.5. Basic block-diagram representation of a continuous-time system.

5.2 Audio CD recording and reproduction system

A practical example of a multi-stage DSP system is the complete recording and playback process of an audio CD. The block diagram below details how physical sound is converted into digital data, recorded, read, and reconstructed back to audio:



Figure 1.6. Audio CD recording and playback as a multi-stage DSP system.

6. Domain comparison: Continuous vs. Discrete

Understanding the dual representations of signals and systems in continuous and discrete domains is foundational to DSP:

Property / Operation	Continuous Time	Discrete Time
Independent Variable	Continuous time variable $t \in \mathbb{R}$	Discrete integer sample index $n \in \mathbb{Z}$
Signal Mathematical Form	$s(t)$ or $x(t)$	$s[n] = s(nT_s)$ or $x[n]$
System Mapping Operator	$y(t) = T\{x(t)\}$	$y[n] = T\{x[n]\}$
Example Operation (Squarer)	$y(t) = x^2(t)$	$y[n] = x^2[n]$

7. Chapter summary

- Signals represent information encoded as physical variations over time or spatial dimensions.
- A single signal can exist across equivalent acoustic, electrical, magnetic, and digital forms.
- Continuous-time signals $s(t)$ are modeled as continuous functions of time t .
- Sampling continuous-time signals at interval T_s yields discrete-time sequences $s[n] = s(n T_s)$.

- Systems operate on input signals to produce new output signals or transformed representations.
- Block diagrams visually map system processing pipelines, mathematical operations, and signal flows.
- Mathematical representations enable precise modeling, interaction analysis, and system design according to exact specifications.

8. Notation

Symbol	Meaning
t	Continuous time variable (independent variable)
n	Discrete sample index (integer, $n \in \mathbb{Z}$)
T_s	Sampling period (time interval between consecutive samples)
$s(t), x(t)$	Continuous-time input signals
$s[n], x[n]$	Sampled discrete-time signals / sequences
$y(t), y[n]$	Continuous-time and discrete-time system output signals
$T\{\cdot\}$	System transformation operator
$\hat{x}(t)$	Reconstructed continuous-time output signal